

## Project 2

# Joint Detection of AI-Generated Images and Post-Processing Alterations in Real-World Scenarios

MSc in Engineering in Computer Science and Artificial Intelligence  
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Lorenzo Sambucci – 2086568

Daniele Valenti – 2085221

Davide Sparvoli – 2048288

# Outline

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- 2 **State of the art** RRDataset, forensic noise features, multi-task learning
- 3 **Proposed method** shared MaxViT-T backbone + two heads, three additions
- 4 **Dataset** balanced subset of RRDataset, with a reasonable split for training, validation and test set
- 5 **Experimental setup** configuration and proposed novelties
- 6 **Model evaluation** baselines vs multi-task comparison, performing Ablation A (components) and Ablation B (loss weighting)
- 7 **Conclusions** results, limitations

# Problem statement

Given a single image, answer to two questions:

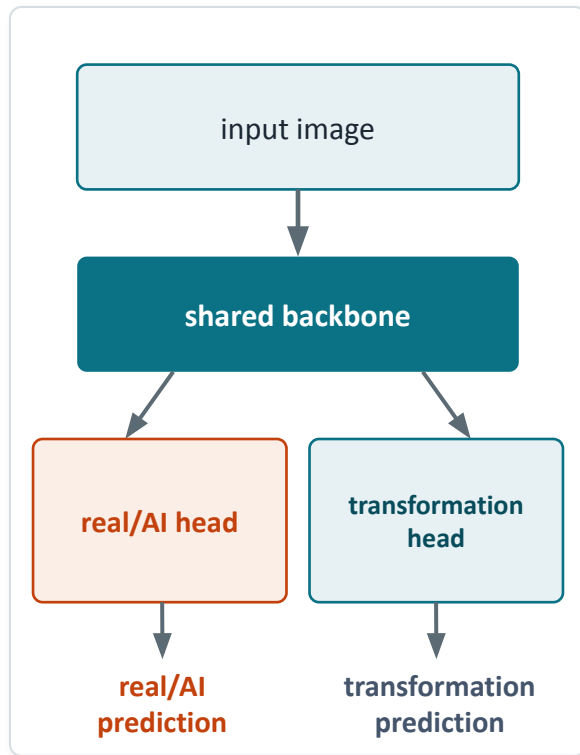
- Is it a real photograph or AI-generated?
- Which post-processing operation was applied? (original, transfer, or redigital)

## Why it is hard

Transmission and re-digitization cover the frequency traces detectors rely on. Performance measured on clean images does not survive the real world, where images are often post-processed.

## Project objectives

- One shared backbone, two heads, joint weighted CE loss
- Real/AI accuracy broken down per transformation and per class
- Ablation on loss weighting: do the tasks compete or complement?



# State of the art

## The Robustness Gap

ICCV '25

*Li et al. — RRDataset*

Li et al. introduced the RRDataset, benchmarking detectors under real-world situations like internet transmission and re-digitization. The study highlights a drop in performance (70-97 accuracy points lost) when moving from clean images to real-world (transmitted, re-digitized) ones .

## Joint architectures

CVPR '23

*Shao et al. DGM4 / HAMMER*

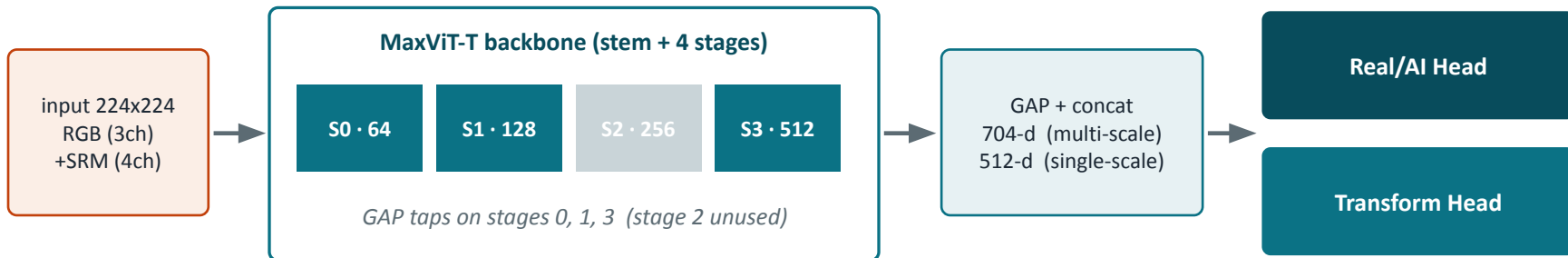
The field is moving towards multi-task reasoning.

For example, Shao et al. proposed DGM4 and the HAMMER framework to jointly detect manipulations and ground tampered areas using a shared representation.

## Our project:

We merge these two paradigms by applying a multi-task architecture to bridge the robustness gap in real-world scenarios.

# Proposed method



**Joint loss:**  $L = w_{\text{fake}} \cdot \text{CE}_{\text{fake}} + w_{\text{tr}} \cdot \text{CE}_{\text{tr}}$

**Learned loss variant (Kendall 2018):**  $L = \sum_i \exp(-s_i) \cdot \text{CE}_i + 1/2 \cdot s_i$

*Proposed novelties (each isolated later in Ablation A / B):*

## Multi-scale features

Concat GAP of stages 0+1+3 (704-d):  
forensic traces are low-level, early stages  
should carry signal the semantic last  
stage abstracts away

## SRM 4th channel

5×5 high-pass residual of the crop; stem  
conv extended to 4 inputs, pretrained  
RGB weights are copied, 4th initialised as  
their mean

## Learned loss weighting

per task uncertainty weights  $s_i$  trained  
jointly with the network

# Dataset: a balanced, leakage-free subset of RRDataset

1

## Download & extract (cached)

RRDataset archive on Zenodo

2

## Group by source image

keep only complete triplets: original + transfer + redigital

3

## Sample 2,000 triplets/class

12000 images in total

4

## Split 70% (train), 15% (validation), 15% (test)

the 3 versions of a photo stay in the same split, to avoid data leakage

### Class balance — verified on disk before training

| split | per (transf. x class) cell | total |
|-------|----------------------------|-------|
| train | $1,400 \times 6$           | 8,400 |
| val   | $300 \times 6$             | 1,800 |
| test  | $300 \times 6$             | 1,800 |

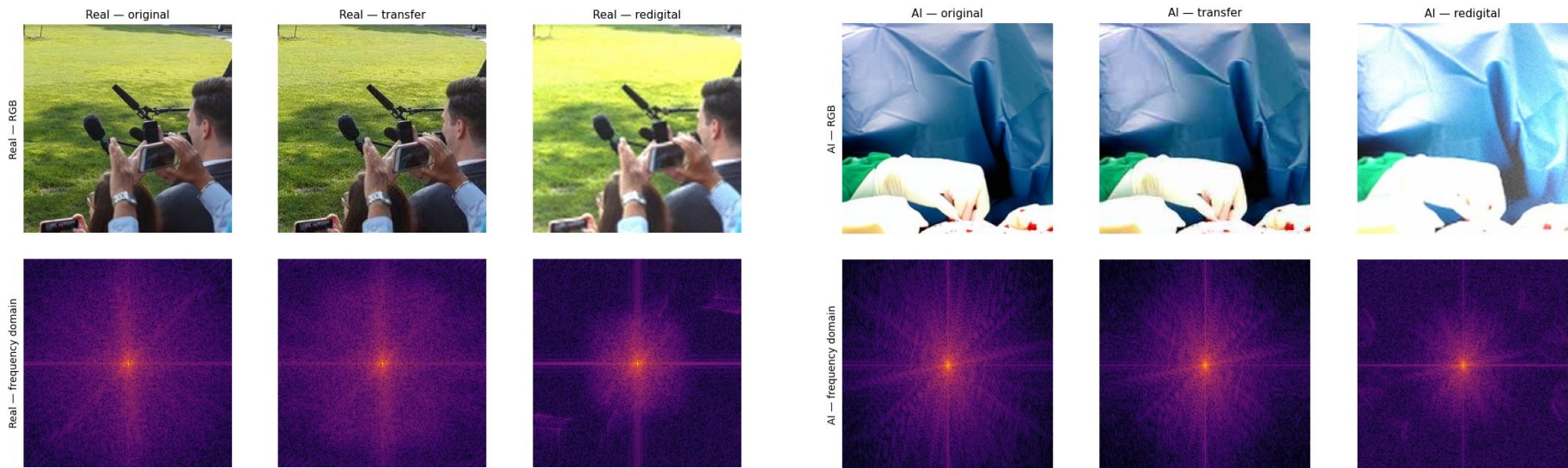
*A dedicated cell re-counts every group and asserts equality, training stops if unbalanced*

### Preprocessing: never resize

Resizing destroys the frequency traces both tasks rely on, instead we use random crop (224) and horizontal flip ( $p=0,5$ ) in training for robustness. We use only center crop (224) in evaluation.

SRM residual computed on the same crop/flip, on  $[0,1]$  pixels before ImageNet normalization.

# Dataset: Frequency analysis



## Our idea:

Frequency content may carry signal beyond RGB, this motivated testing a dedicated frequency channel (SRM)

# Experimental setup

## Configuration

|                         |                                                                           |
|-------------------------|---------------------------------------------------------------------------|
| <b>Backbone</b>         | MaxViT-T, pretrained on ImageNet                                          |
| <b>Training</b>         | Full fine-tuning: backbone LR x 0.1                                       |
| <b>Optimizer</b>        | AdamW: LR 1e-3                                                            |
| <b>Batch / epochs</b>   | 32 / 10 (fixed)                                                           |
| <b>Trainable params</b> | About 30.9 M                                                              |
| <b>Seed</b>             | 9999                                                                      |
| <b>Hardware</b>         | GPU: RTX 5060 Ti 16 GB<br>CPU: AMD Ryzen 5600x<br>RAM: 64 GB DDR4 3600MHz |

## Experiment map:

**1a - 1b: unimodal baselines**

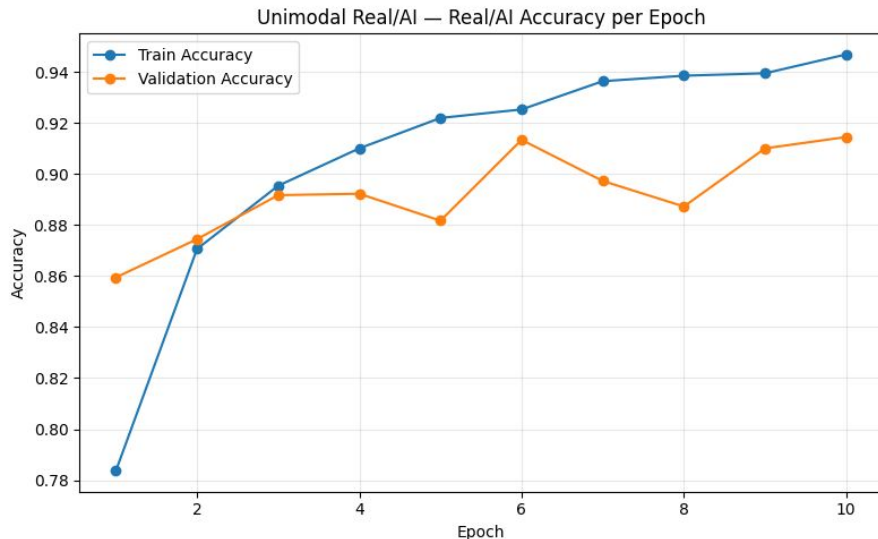
**2a - 2b: multi-task**

**Ablation A: components**

**Ablation B: loss weights**

**ResNet18 cross-backbone comparison**

# Model evaluation: Unimodal Real/AI task

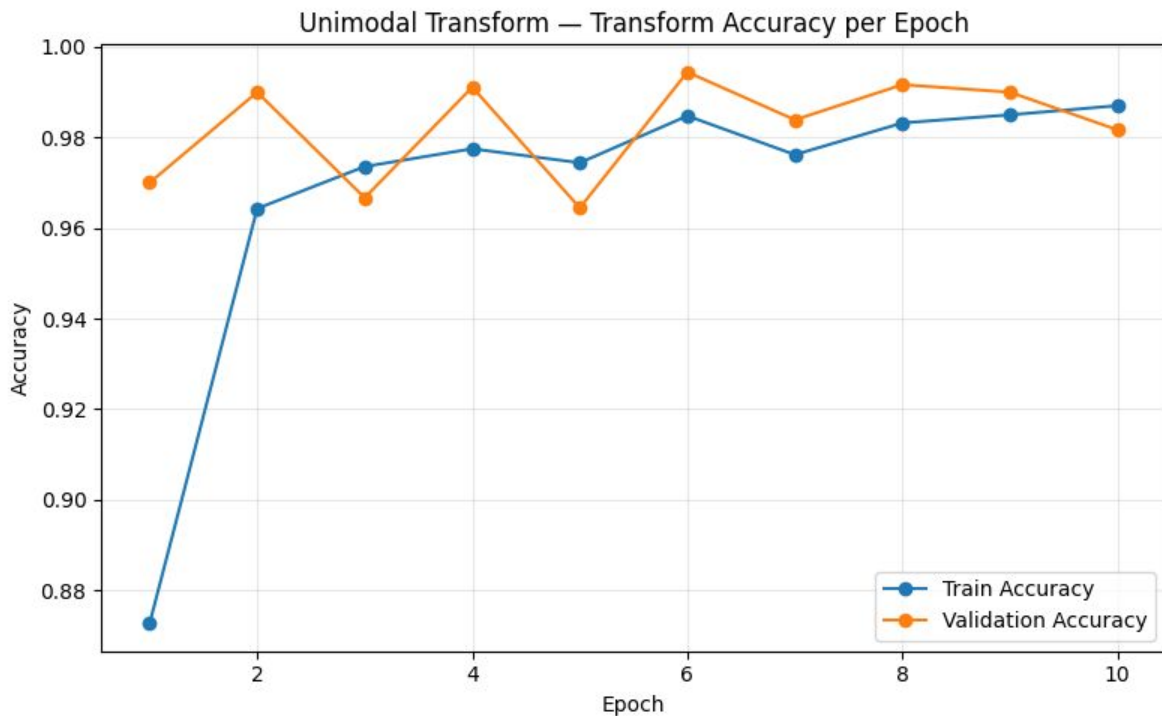


## Breakdown

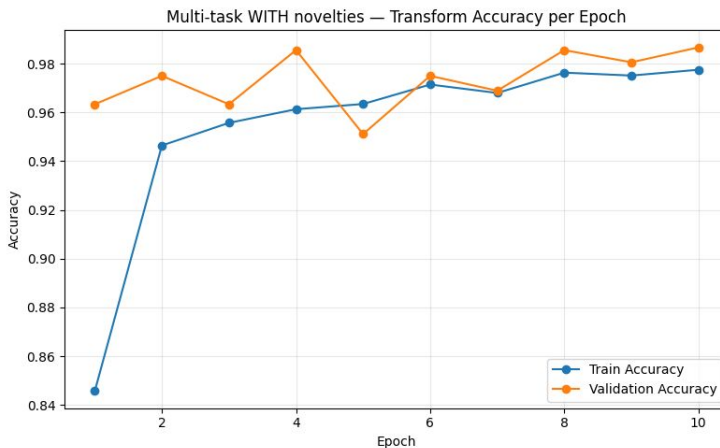
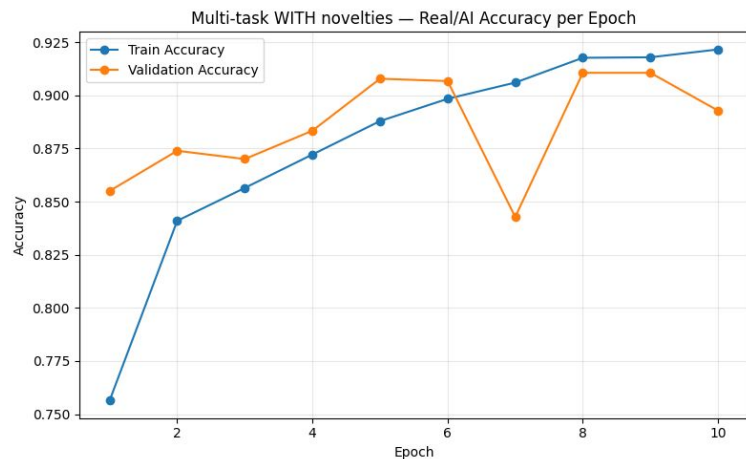
The breakdown below shows where the model struggles the most. The model recognizes quite well AI images, no matter the transformation, but struggles in particular when a real image is re-digitized or transferred.

| Transformation | Total Accuracy | Accuracy (Real) | Accuracy (AI) |
|----------------|----------------|-----------------|---------------|
| Original       | 0.9433         | 0.917           | 0.970         |
| Transfer       | 0.8983         | 0.847           | 0.950         |
| Redigital      | 0.8967         | 0.887           | 0.907         |

# Model evaluation Unimodal Transform task

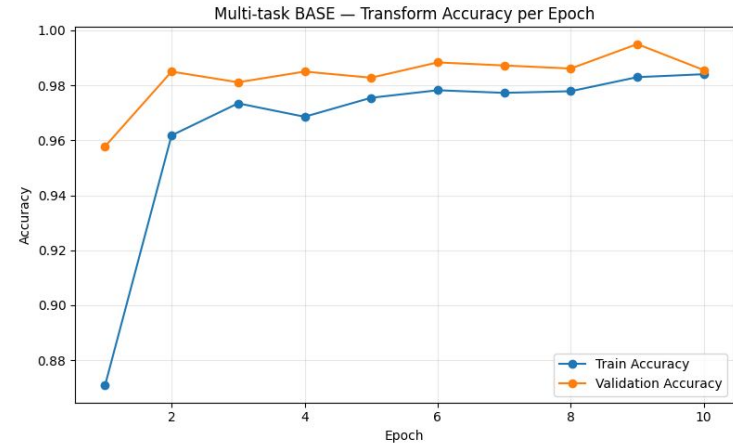
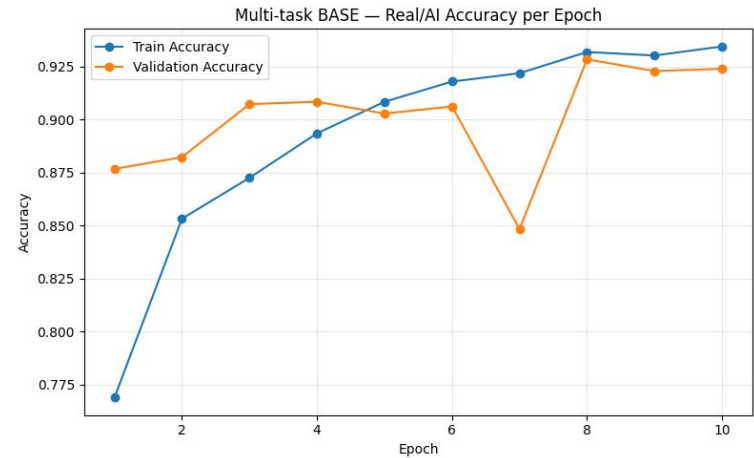


# Model evaluation: Multi-task with novelties



| Transformation | Total Accuracy | Accuracy (Real) | Accuracy (AI) |
|----------------|----------------|-----------------|---------------|
| Original       | 0.9450         | 0.930           | 0.960         |
| Transfer       | 0.8600         | 0.747           | 0.973         |
| Redigital      | 0.8717         | 0.810           | 0.933         |

# Model evaluation: Multi-task base



| Transforma<br>tion | Total<br>Accuracy | Accuracy<br>(Real) | Accuracy<br>(AI) |
|--------------------|-------------------|--------------------|------------------|
| Original           | 0.9483            | 0.933              | 0.963            |
| Transfer           | 0.8883            | 0.870              | 0.907            |
| Redigital          | 0.8667            | 0.860              | 0.873            |

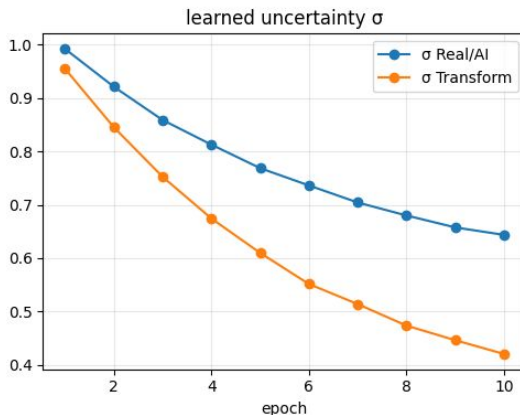
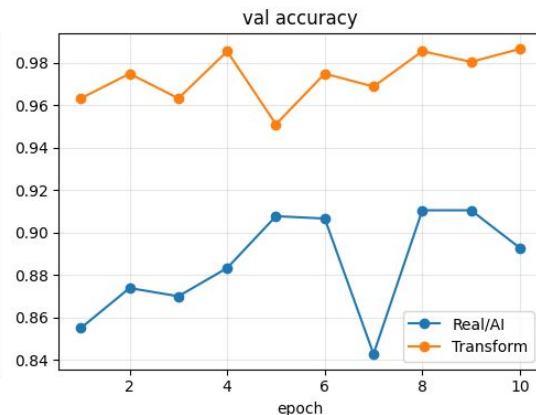
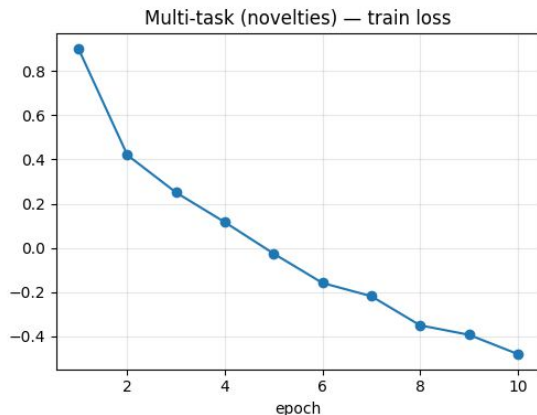
# Model evaluation: comparison between multitask and baselines

Joint-training effect:

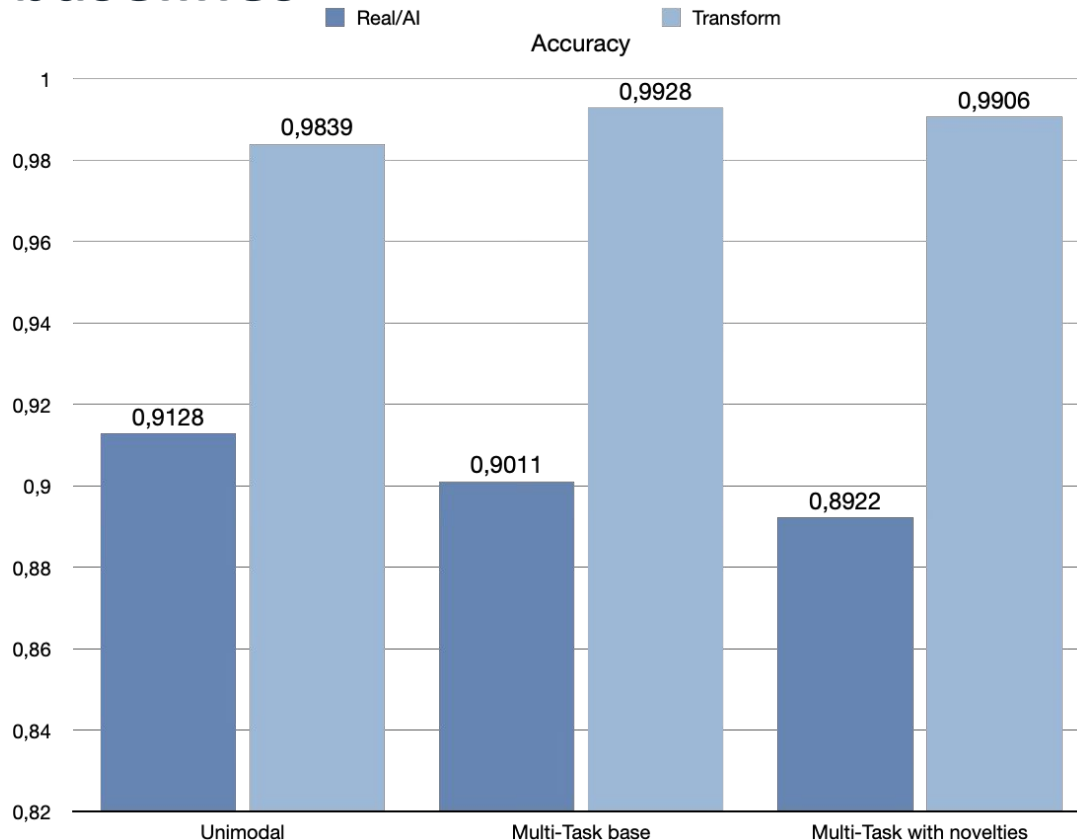
|           | Unimodal baselines | Multi-task base |
|-----------|--------------------|-----------------|
| Real/AI   | 0.9128             | 0.9011          |
| Transform | 0.9839             | 0.9928          |

Novelties effect:

|           | Multi-task base | Multi-task with novelties |
|-----------|-----------------|---------------------------|
| Real/AI   | 0.9011          | 0.8922                    |
| Transform | 0.9928          | 0.9906                    |



# Model evaluation: comparison between multitask and baselines



**Joint training is asymmetric**

**Transform: 0.9839 to 0.9928**

**Real/AI: 0.9128 to 0.9011**

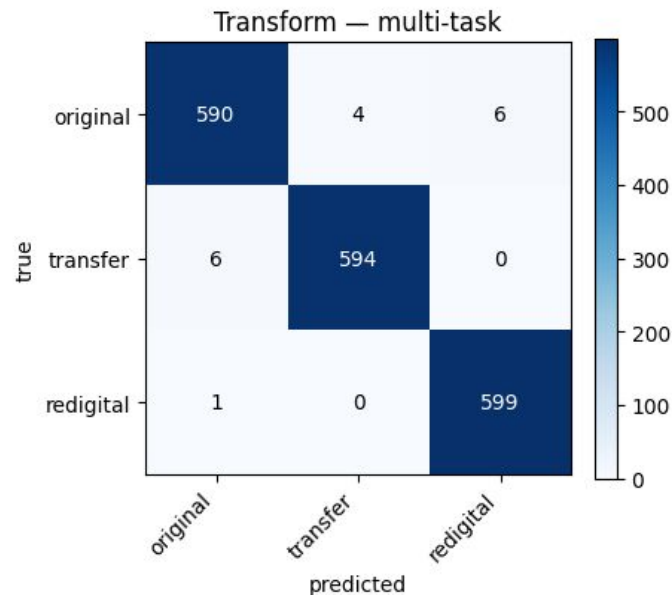
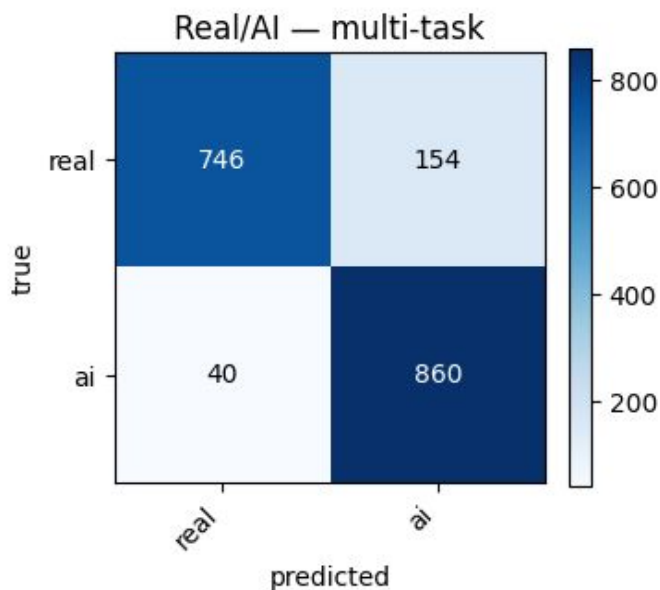
the shared backbone favors the Transform task

**Novelties do worse!**

**Transform: 0.9928 to 0.9906**

**Real/AI: 0.9011 to 0.8922**

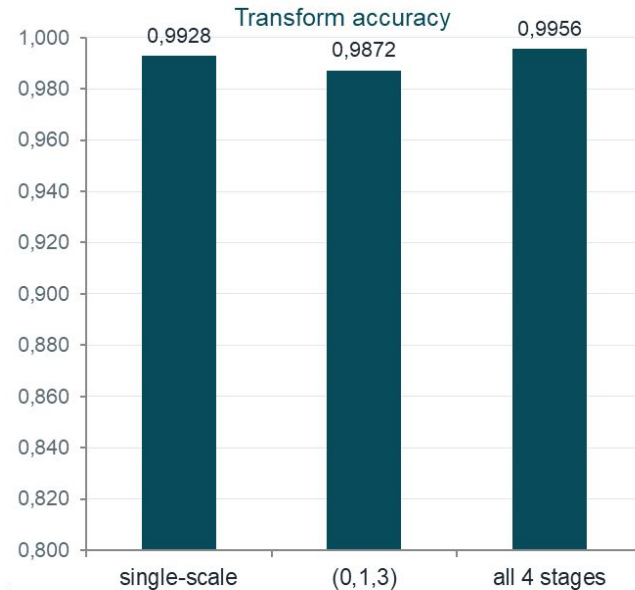
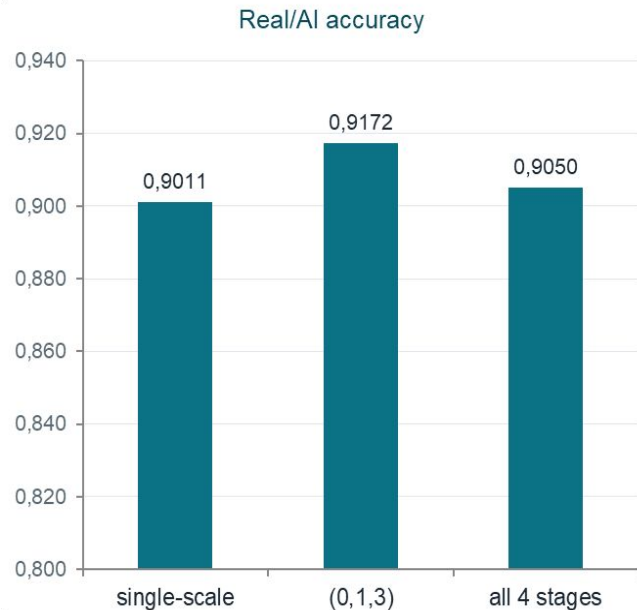
# Model evaluation: focus on Multi-task with novelties



## Important findings:

1. At epoch 7, the model processes a batch that is particularly challenging for it, recording a significant drop in accuracy for the Real/AI task.
2. The model classifies the images into their respective classes quite well, showing a slight weakness in classifying real images as AI generated.

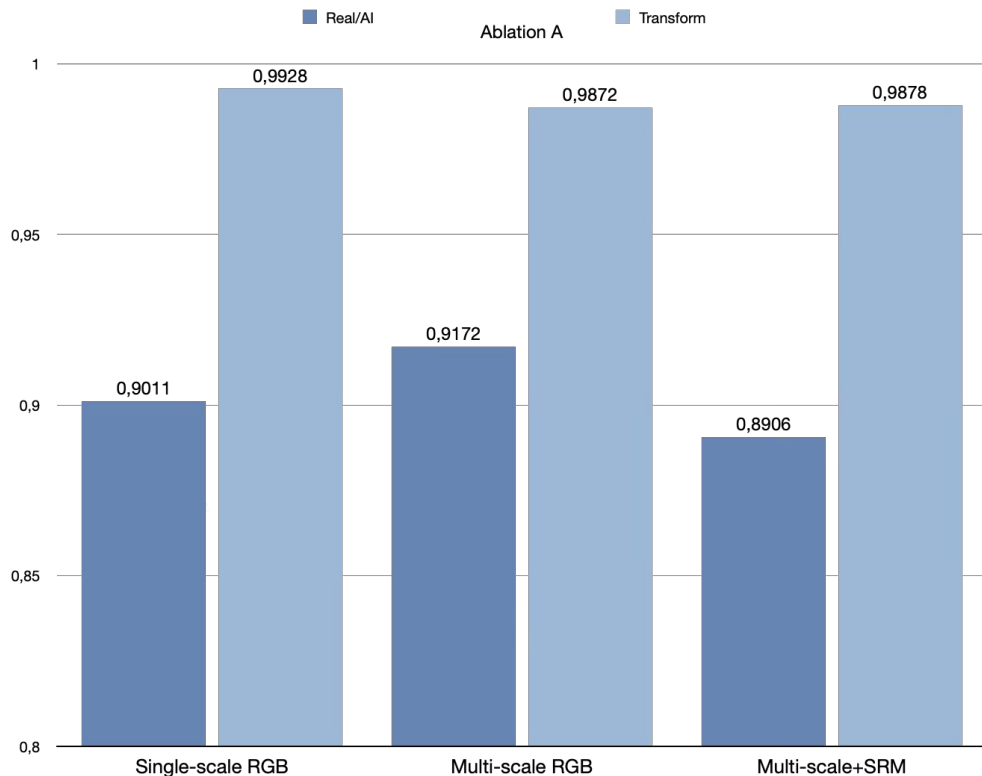
# Model evaluation: Stage 2 MaxVit-T



## The trade off:

Including stage 2 (middle-level features), slightly enhances the Transform task accuracy, but lowers the Real/AI one. Since the Real/AI task is the one that struggles the most, we opted for multi-scale with 0,1,3 stages as a trade off.

# Model evaluation: Ablation A



## Setting:

Ablation A isolates the novelties by performing 3 experiments, using fixed 1.0/1.0 weights for the loss:

- Single-scale RGB
- Multi-scale RGB
- Multi-scale+SRM

## Multi-scale helps:

0.9172 is the best Real/AI result of the whole project and it is above the unimodal baseline (0.9128).

## SRM degrades the performance:

When using SRM, the multi-task model loses points.

# Model evaluation: Ablation A

**Single-scale RGB:**

| Transformation | Total Accuracy | Accuracy (Real) | Accuracy (AI) |
|----------------|----------------|-----------------|---------------|
| Original       | 0.9483         | 0.933           | 0.963         |
| Transfer       | 0.8883         | 0.870           | 0.907         |
| Redigital      | 0.8667         | 0.860           | 0.873         |

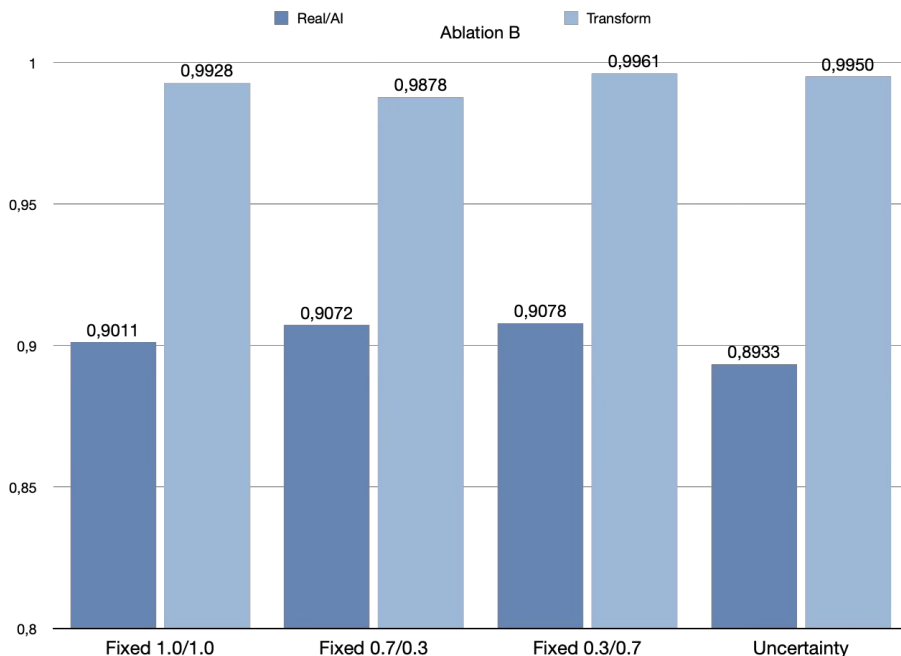
**Multi-scale RGB:**

| Transformation | Total Accuracy | Accuracy (Real) | Accuracy (AI) |
|----------------|----------------|-----------------|---------------|
| Original       | 0.9567         | 0.933           | 0.980         |
| Transfer       | 0.8983         | 0.857           | 0.940         |
| Redigital      | 0.8967         | 0.877           | 0.917         |

**Multi-scale + SRM:**

| Transformation | Total Accuracy | Accuracy (Real) | Accuracy (AI) |
|----------------|----------------|-----------------|---------------|
| Original       | 0.9333         | 0.900           | 0.967         |
| Transfer       | 0.8750         | 0.787           | 0.963         |
| Redigital      | 0.8633         | 0.787           | 0.940         |

# Model evaluation: Ablation B



## Setting:

Ablation B evaluates the loss functions, by performing 4 experiments using the basic single-scale RGB architecture:

- Fixed weights 1.0/1.0
- Fixed weights 0.7/0.3
- Fixed weights 0.3/0.7
- Learned Uncertainty

## Fixed 0.3/0.7 wins:

These weights yield the best performance result. Uncertainty loss loses because the weight of the easier task (Transform) ends up dominating the loss.

# Model evaluation: Ablation B

Fixed weights at 1.0/1.0:

| Transformation | Total Accuracy | Accuracy (Real) | Accuracy (AI) |
|----------------|----------------|-----------------|---------------|
| Original       | 0.9483         | 0.933           | 0.963         |
| Transfer       | 0.8883         | 0.870           | 0.907         |
| Redigital      | 0.8667         | 0.860           | 0.873         |

Fixed weights at 0.7/0.3:

| Transformation | Total Accuracy | Accuracy (Real) | Accuracy (AI) |
|----------------|----------------|-----------------|---------------|
| Original       | 0.9517         | 0.943           | 0.960         |
| Transfer       | 0.8933         | 0.853           | 0.933         |
| Redigital      | 0.8767         | 0.830           | 0.923         |

Fixed weights at 0.3/0.7:

| Transformation | Total Accuracy | Accuracy (Real) | Accuracy (AI) |
|----------------|----------------|-----------------|---------------|
| Original       | 0.9583         | 0.960           | 0.957         |
| Transfer       | 0.8850         | 0.853           | 0.917         |
| Redigital      | 0.8800         | 0.843           | 0.917         |

Uncertainty loss:

| Transformation | Total Accuracy | Accuracy (Real) | Accuracy (AI) |
|----------------|----------------|-----------------|---------------|
| Original       | 0.9517         | 0.963           | 0.940         |
| Transfer       | 0.8867         | 0.877           | 0.897         |
| Redigital      | 0.8417         | 0.780           | 0.903         |

# Model evaluation: Best of both

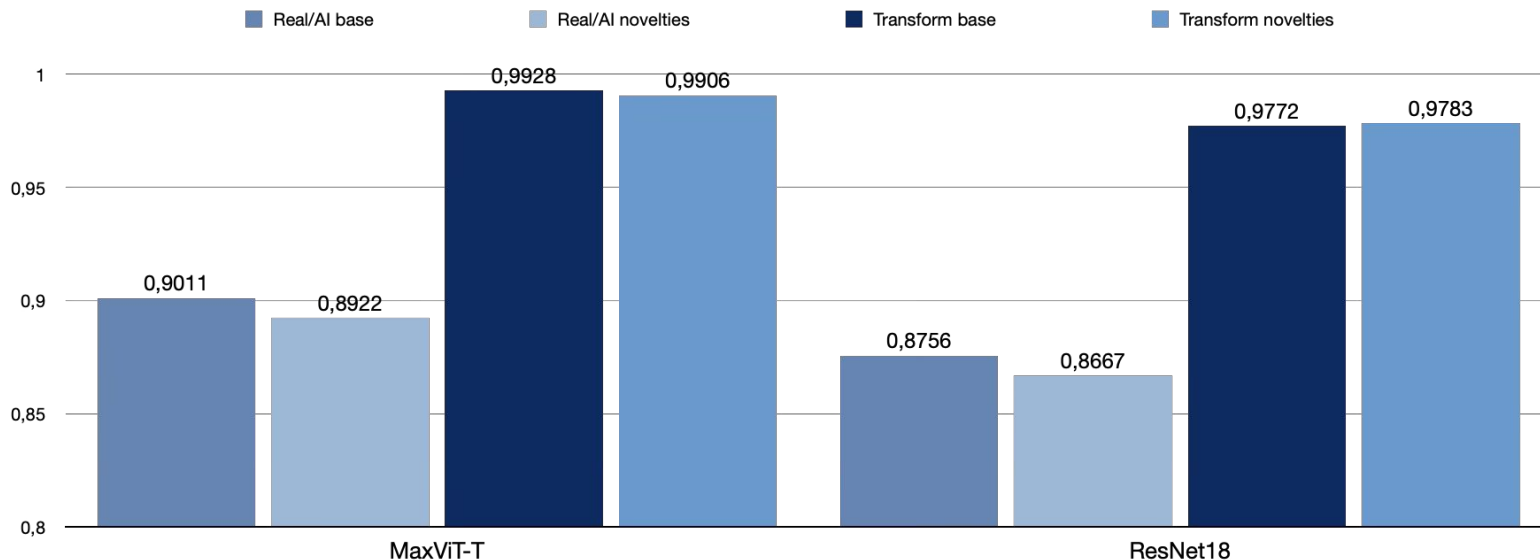
| Transformation | Total Accuracy | Accuracy (Real) | Accuracy (AI) |
|----------------|----------------|-----------------|---------------|
| Original       | 0.9450         | 0.933           | 0.957         |
| Transfer       | 0.8933         | 0.857           | 0.930         |
| Redigital      | 0.8950         | 0.883           | 0.907         |

## Best of both:

The two ablations we performed highlighted the possible best configuration: Multi-scale RGB (no SRM) + fixed 0.3/0.7 weights.

In this setting, Real/AI accuracy is 0.9111 and Transform accuracy is 0.9889 (below each component's individual best (0.9172/0.9961). This shows that the improvement are not independent, but the resulting model is a robust one.

# Cross backbone check: ResNet 18



**Motivation:** since the novelties do not improve the performance on the multi-task model, which uses MaxViT-T as the backbone, we performed a test: apply these novelties to an older network, ResNet 18.

**Result:** same direction as MaxViT, the proposed novelties do not improve the accuracy.

# Conclusions

## Key findings:

**Asymmetric Joint Training:** A shared backbone favors the easier transformation task while penalizing Real/AI detection.

**Feature Aggregation:** Multi-scale implementations achieved our best Real/AI accuracy (0.9172).

**Loss Optimization:** Fixed asymmetric weighting (0.3/0.7) dominates equal weights. Learned uncertainty collapses because it favors the simpler task (transform).

## Future work

- Better understand why our implementation of SRM is counter productive, to work on a possible fix that actually enhances performance.
- The redigital folder of our dataset contains more than one type of re-digitization (e.g. scan of a color printout, photo of a color printout, photo of an image displayed on a screen); further labeling might yield better results.

# References

- 1 Li, Chunxiao, et al. "Bridging the Gap Between Ideal and Real-world Evaluation: Benchmarking AI-Generated Image Detection in Challenging Scenarios." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2025.
- 2 Shao, Rui, Tianxing Wu, and Ziwei Liu. "Detecting and grounding multi-modal media manipulation." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2023.
- 3 Kendall, A., Gal, Y., Cipolla, R. "Multi-Task Learning Using Uncertainty to Weigh Losses for Scene Geometry and Semantics." CVPR 2018.
- 4 Tu, Z., et al. "MaxViT: Multi-Axis Vision Transformer." ECCV 2022.
- 5 Zhou, P., et al. "Learning Rich Features for Image Manipulation Detection." CVPR 2018.

**Dataset:** <https://zenodo.org/records/14963880>    Code: project GitHub repository